

Student Marks Analyzer AI Agent

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1. Abstract

The Student Marks Analyzer AI Agent is an intelligent system designed to analyze student academic performance automatically.

The system takes student marks as input, calculates the average, classifies performance into Low, Medium, or High, and provides suitable suggestions.

This project reduces manual effort and helps teachers and students to understand academic performance efficiently.

The system works autonomously using rule-based decision logic, which makes it a simple and effective AI Agent.

2. Introduction

Artificial Intelligence plays a vital role in automating decision-making processes.

In educational institutions, analyzing student performance manually is time-consuming.

The Student Marks Analyzer AI Agent automates this process by analyzing marks and providing intelligent feedback.

This project demonstrates how AI Agents can be applied in the education domain.

3. Problem Statement

In the existing system, teachers manually calculate marks and evaluate student performance.

This process is inefficient and prone to errors when handling a large number of students. Hence, there is a need for an automated AI-based system to analyze student marks and generate performance reports.

4. Objectives of the Project

To collect student marks as input

To calculate average marks

To classify performance levels

To provide intelligent suggestions

To reduce manual effort

To function as an autonomous AI Agent

5. Scope of the Project

Applicable for school and college students

Uses a limited number of subjects

Rule-based analysis

Educational purpose only

Can be enhanced in future with ML models

6. System Architecture

The system consists of four main modules:

1. Input Module
2. Processing Module
3. Decision-Making Module
4. Output Module

These modules work together to analyze student marks and generate intelligent output.

7. Algorithm

1. Start
2. Read student name
3. Read subject marks
4. Calculate average marks
5. Compare average with thresholds
6. Classify performance

7. Generate suggestion

8. Display result

9. Stop

8. Implementation

The project is implemented using Python programming language in Google Colab.
The AI Agent uses rule-based logic to analyze student marks and make decisions automatically.

Technologies Used:

Python

Google Colab

```
student_name = input("Enter Student Name: ")

print("Student Name:", student_name)

maths = int(input("Enter Maths Marks: "))
science = int(input("Enter Science Marks: "))
english = int(input("Enter English Marks: "))

print("Maths:", maths)
print("Science:", science)
print("English:", english)

*** Enter Student Name: Yoga
Student Name: Yoga
Enter Maths Marks: 98
Enter Science Marks: 99
Enter English Marks: 88
Maths: 98
Science: 99
English: 88
```

9. Result

The system successfully analyzes student marks and displays the performance level along with suggestions.

The output is clear and useful for academic evaluation.

Sample Output:

Student Name: Yoga

Average Marks: 95

Performance Level: Good

Agent Suggestion: Get full marks

10. Conclusion

The Student Marks Analyzer AI Agent effectively demonstrates the use of artificial intelligence in academic performance analysis.

The system operates autonomously and reduces manual workload.

This project proves that AI Agents can be efficiently used in educational applications.

11. Future Enhancements

Add more subjects

Integrate machine learning models

Store data in database

Develop web or mobile application

12. References

Python Documentation

AI Concepts and Notes

Online Learning Resources